

Predicting Acid Dissociation from UMA Embeddings

A frozen foundation model that never saw a pK_a value — and what a new solvent actually costs

Srikanth Mohan · MIT Undergraduate Research Technology Conference · github.com/SMLion1959422/PKa-calculator-Frozen-UMA-

1 · Why pK_a , why now

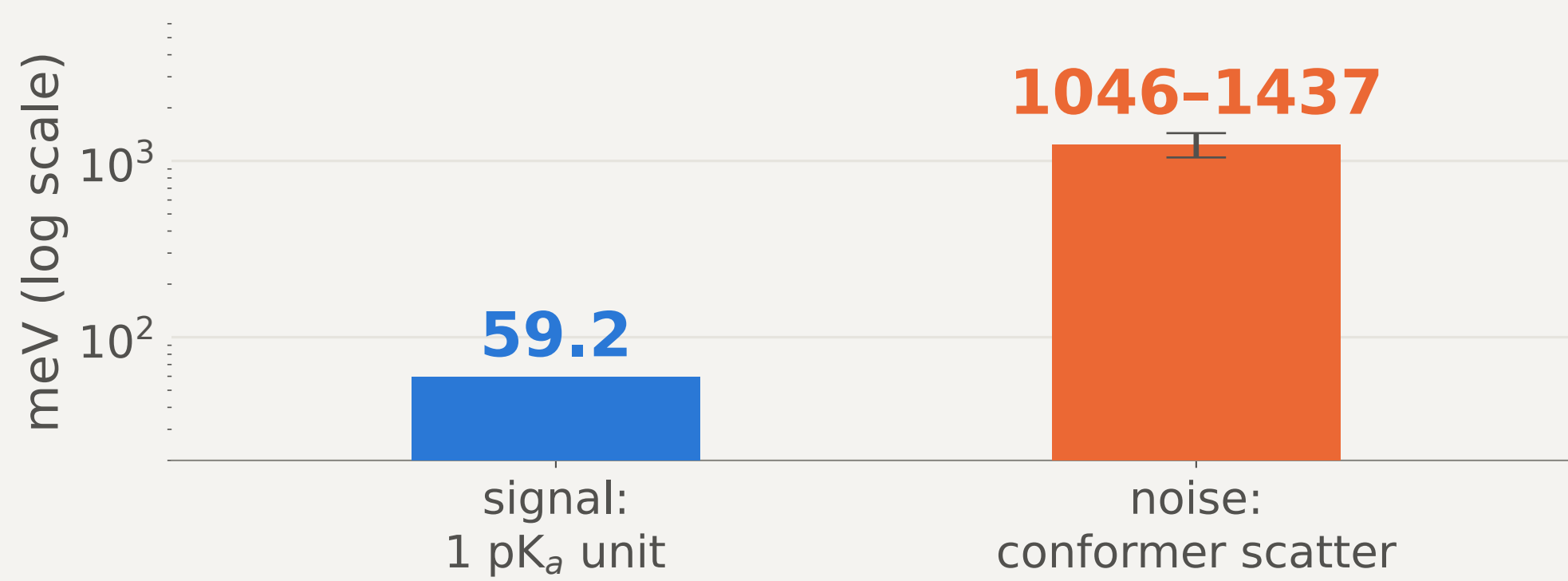
Bringing a drug to market costs billions, mostly on candidates that fail. Many failures trace to physicochemical causes knowable before synthesis.

Those properties depend on which ionization state a molecule adopts — set by its pK_a .

Goal: a free, offline, license-free predictor built on a foundation model never trained on pK_a data.

2 · UMA's energies cannot work

The direct route — compute the deprotonation energy and convert — fails for a reason that is arithmetic, not tuning.



The noise is 18–24× the entire signal. GFN2-xTB, an established semi-empirical method, fails identically — a precision wall shared across methods, not a flaw in UMA.

3 · Use the representations instead

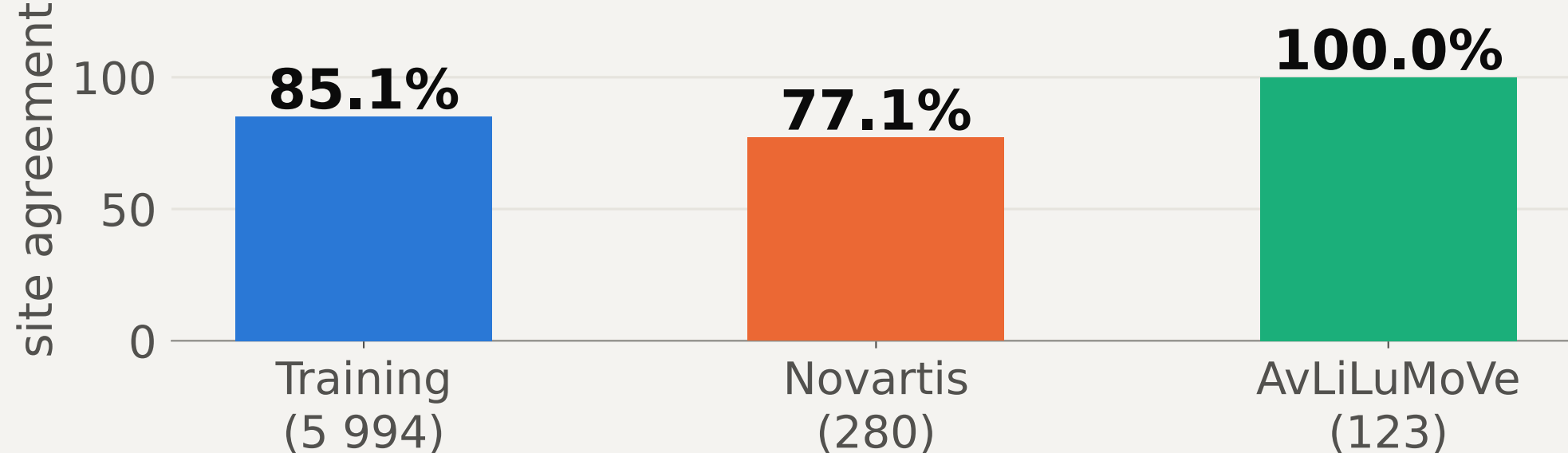
UMA stays frozen. Instead of its scalar energy we read the 128-dim per-atom embeddings from the input to its energy head.

Critically we pool them AROUND THE IONIZABLE SITE, not over the whole molecule: pK_a is local, and global averaging dilutes it as molecules grow.

[h_{prot} ; h_{deprot} ; $h_{prot} - h_{deprot}$; ΔE ; n_{heavy}] $\rightarrow$ gradient-boosted regressor. UMA's weights are never updated.

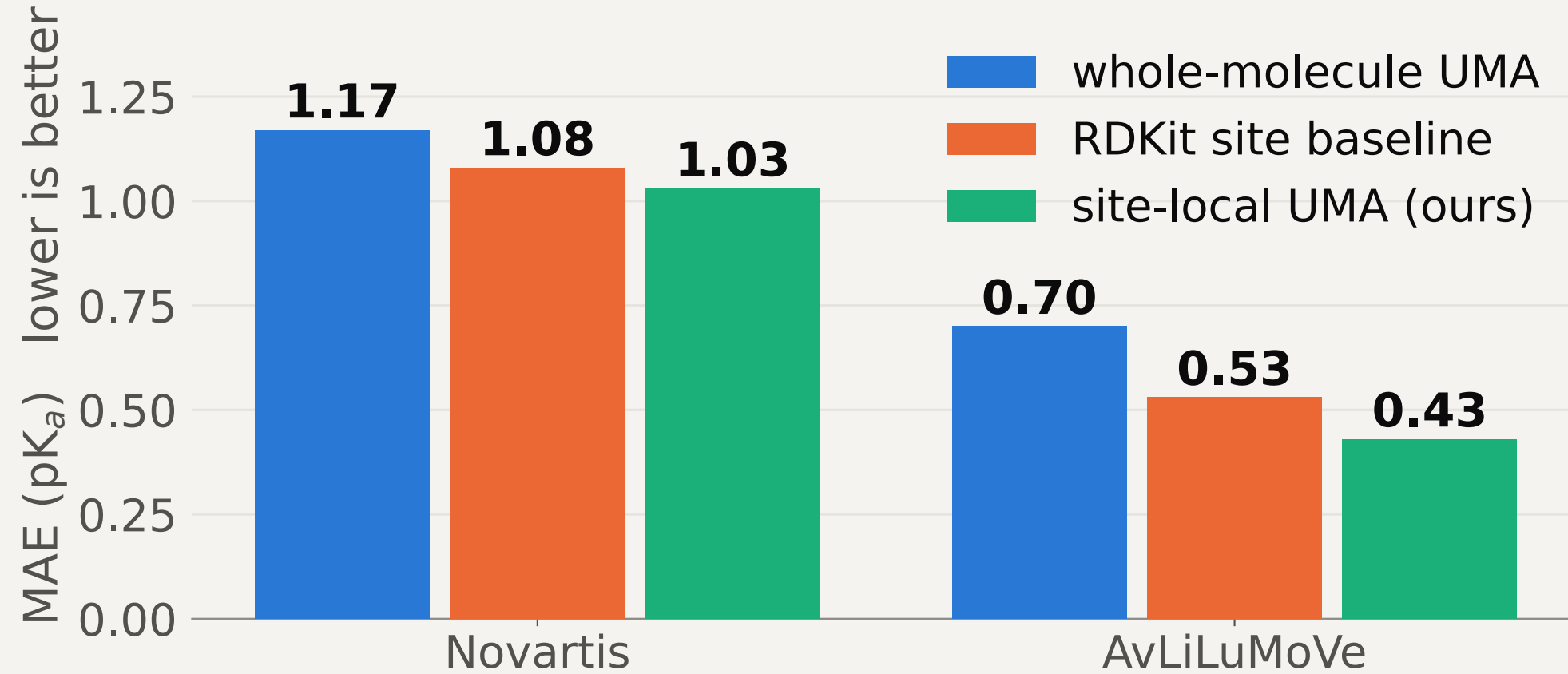
4 · Site assignment dominates

The standard SMARTS rule picks a DIFFERENT atom from the dataset's own annotation for 23% of the Novartis benchmark.



Anyone benchmarking on Novartis is partly measuring site detection. We also found tetrazoles could never be deprotonated — the SMARTS pointed at the ring carbon.

5 · Aqueous accuracy



Site-local pooling beats both the original whole-molecule UMA model and a same-features RDKit fingerprint baseline, on identical splits.

6 · What the embeddings add

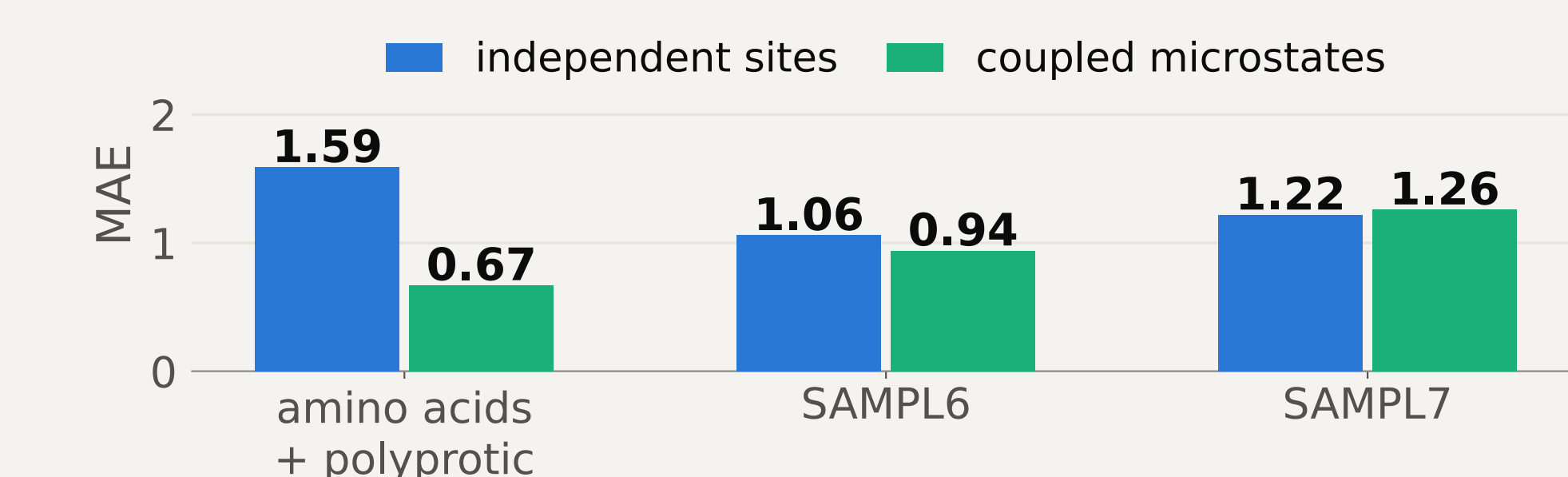
Given the CORRECT site, site-local UMA reaches 0.86 on Novartis. The gap up to 1.03 is automatic site selection — not the model.

Scaffold-split CV: 0.72 — no shared Bemis-Murcko core

UMA beats the fingerprint baseline by 13% at the annotated site and 18% on AvLiLuMoVe — the representation carries real information.

7 · One pK_a is not enough

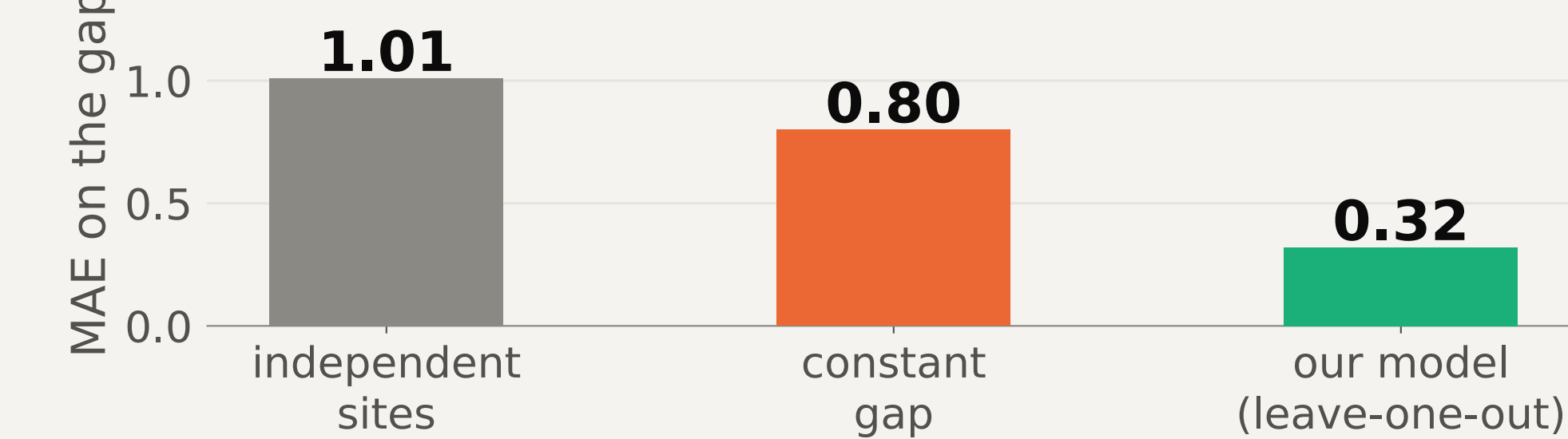
Real molecules have several interacting ionizable sites. Scoring them independently is wrong by construction — it is why amino acids failed.



Amino acids 1.59 $\rightarrow$ 0.67 ($p = 4 \times 10^{-7}$, better on 24/26). pl error 0.49 $\rightarrow$ 0.29; net charge at pH 7.4 correct for 25/26.

8 · Coupling validated, no ML

For a symmetric diacid the intrinsic pK_a cancels exactly: $pK_{a2} - pK_{a1} = \log_{10} 4 + W$ — so the one fitted constant is testable with no ML involved.



17 diacids / diamines. Held-out glycine: predicted pl 6.06 vs 6.06 experimental — the zwitterion is recovered.

9 · What does a NEW solvent cost?

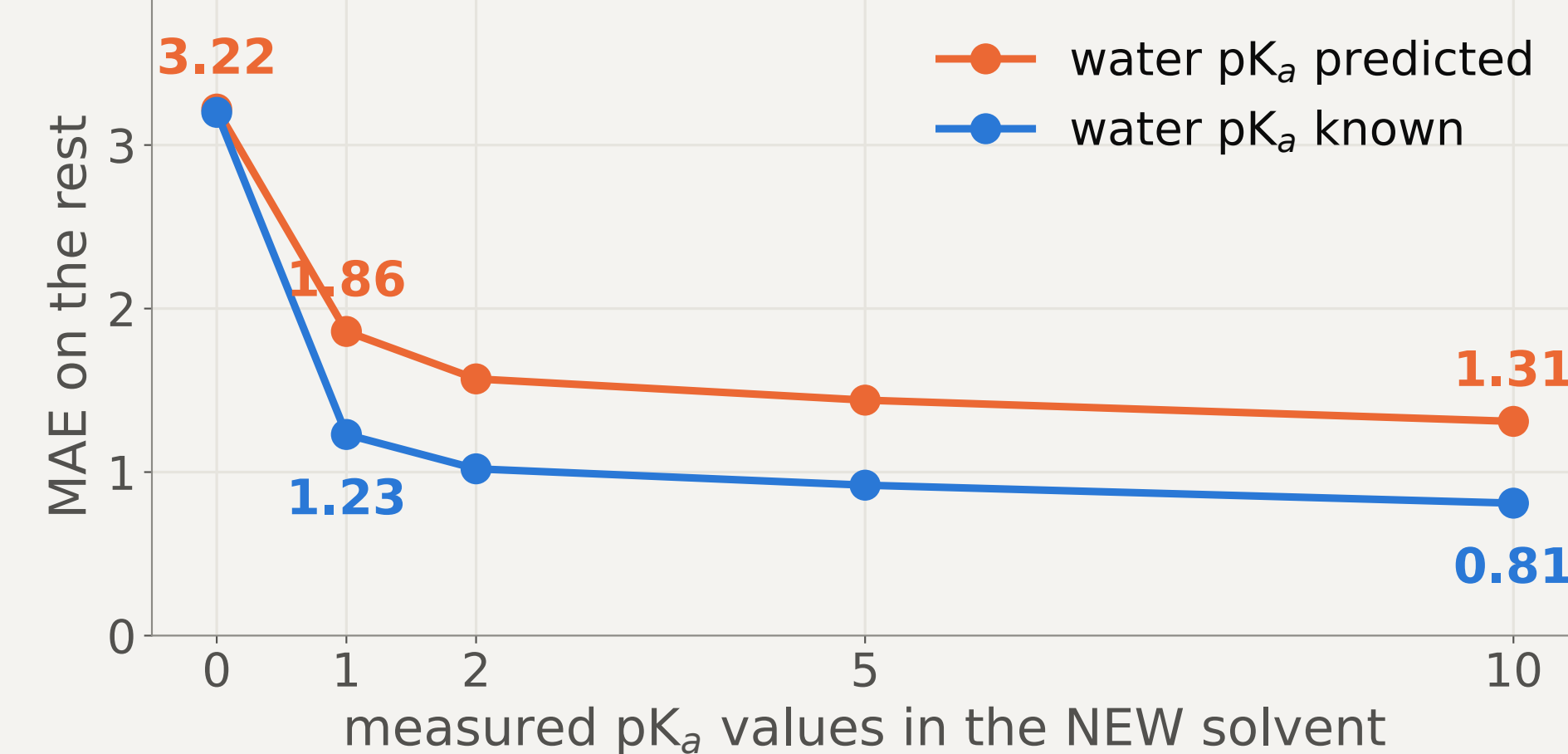
Every published multi-solvent pK_a model reports accuracy on unseen MOLECULES. We could not find one that reports unseen SOLVENTS.

So we measured it. Leave-one-solvent-out: train on seven solvents, predict the eighth, for each in turn.

MAE 3 - 5

Models do not transfer to a solvent they have never seen.

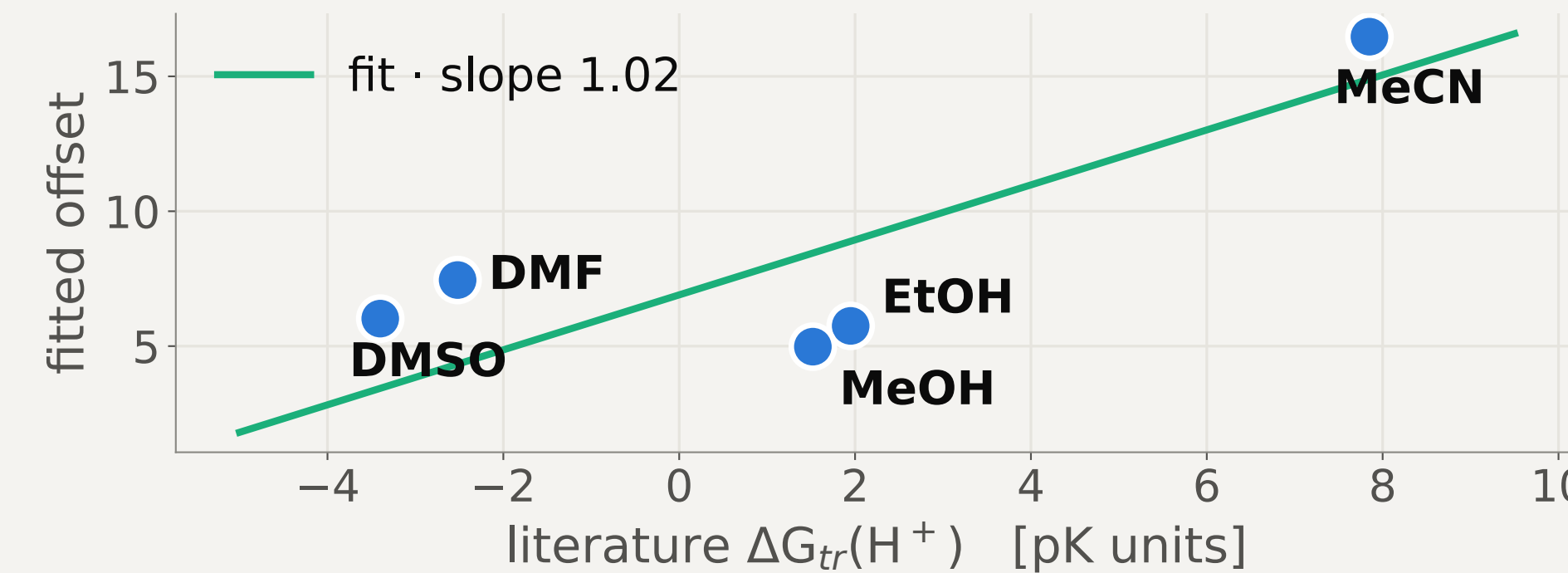
10 · One measurement buys it back



A single measurement removes 40–60% of the error. Mean over held-out solvents, 20 random draws each.

11 · Why: one physical constant

Relative to water the thermodynamic cycle splits the shift into a solute term and ONE constant per solvent: the proton transfer energy.



offset = $1.02 \cdot \Delta G_{tr}(H^+) - 6.66 \cdot \Delta \alpha + 1.95$ LOSO error 0.40

The proton coefficient lands at 1.00 within error, 95% CI [0.84, 1.20] — what the cycle mandates, not something the fit was told.

12 · The rest is anion desolvation

The remainder tracks the solvent's H-bond DONOR strength α — an anion needs donors. It shows NO relation to β , the acceptor scale ($r^2 = 0.03$): the specificity check the physics demands.

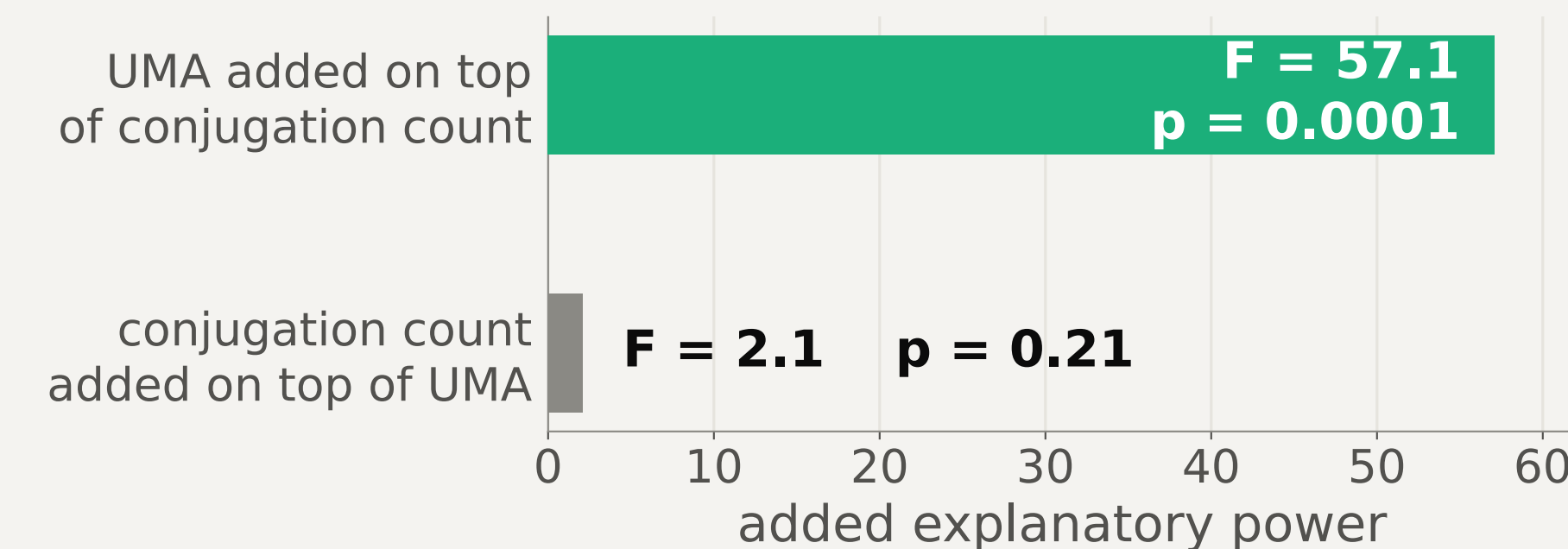
Correcting one bad literature value (DMF) cut leave-one-out error 1.14 $\rightarrow$ 0.40 without adding a single parameter.

13 · What UMA actually knows

If the leftover is anion desolvation, sensitivity to α should depend on how DELOCALIZED the anion's charge is.

UMA probes this directly: deprotonation changes its per-atom embeddings, and WHERE it changes them is where the charge went.

$w_a = ||h(A^-)_a - h(HA)_a||$, summarized by its spread from the site



UMA SUBSUMES the trivial baseline: it adds over a conjugated-atom count, but the count adds nothing over UMA. It also adds over Gasteiger charges and the count combined ($F = 34.5$).

14 · Every alternative, excluded

Just the family label?	tested WITHIN families — survives
Just repeated rows?	molecule-level permutation — survives
Just counting conjugation?	asymmetric subsumption — survives
Just Gasteiger charges?	complementary, $r = +0.23$ — survives
Multiple comparisons?	Holm 23/42, Bonferroni 18/42 — survives

Carboxylic acids are the one exception — chemically expected, since carboxylate delocalization barely varies with substituent.

15 · Honest limitations

- The solvent decomposition rests on 5 solvents and 3 parameters.
- $\Delta G_{tr}(H^+)$ values come from secondary sources cross-checked against each other, not read off a primary table.
- Non-aqueous MAE 0.711 is for solvents PRESENT in training; unseen solvents are 3–5. Different claims.
- Site selection, not the model, still caps aqueous accuracy.
- No cationic-acid data here, so the charge-type prediction is untested.

16 · Conclusions

A frozen foundation model, never trained on pK_a , supplies representations that predict it — but only when read at the ionizable site.

No pK_a model transfers to an unseen solvent. That gap is one physical constant plus anion desolvation — and one measurement recovers most of it.

Matters most where data is sparse: chelators, PFAS, natural products.